



Figure 1: MinnowBoard

<http://dominion.thruhere.net/koen/angstrom/minnow/Angstrom-development-GNOME-image-eglibc-ipk-v2012.12-minnow-2013.05.24.img.xz>

At the time of writing this article, the latest image available for download was 'Angstrom-development-GNOME-image-eglibc-ipk-v2012.12-minnow-2013.05.24.img.xz'

Preparing the microSD card

For the purpose of this article, I am using Ubuntu 12.10 running on a 64 bit machine. Now insert your microSD card into the computer but make sure that it is not mounted. Now, identify the correct raw device name (like /dev/sde - not /dev/sde1). You can do this by using a tool called disk utility in Ubuntu. Now navigate to the folder where you have downloaded the image using the terminal. Type the following command into the terminal window to write the image to the microSD card:

```
xzcat Angstrom-development-GNOME-image-eglibc-ipk-v2012.12-minnow-2013.05.24.img.xz | sudo dd if=/dev/sdx
```

Here, 'sdx' stands for the raw device ID of the microSD card. Please be patient as this operation will take some time. Once successful, remove the microSD card.

Booting MinnowBoard

Now, plug in the microSD card into the MinnowBoard. To power it up, connect the 5V/2.5A power supply to the MinnowBoard. Then, you can either choose to take the GUI or the Command Line approach. In the GUI approach, you will connect an external display to the DVI port on MinnowBoard, and the USB compatible keyboard and mouse on the corresponding USB host ports.

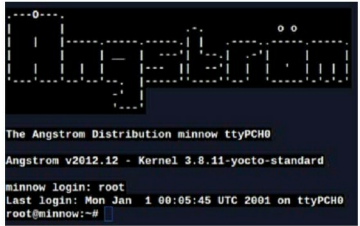


Figure 2: Angstrom



Figure 3: USER LED(s) on MinnowBoard

In the Command Line approach, connect the MinnowBoard to your computer using the USB cable provided. The next step is to setup the terminal emulator so that we can access the board's serial console. There are two possible ways here. Either use a terminal emulator or the Linux 'screen' command. Let us discuss the first method. Install minicom/Gtkterm in Ubuntu. I personally prefer minicom. Install it by typing the following command in the terminal window:

```
sudo apt-get install minicom
```

Start minicom by typing the following code into the terminal:

```
sudo minicom -s
```

Configure it by selecting the correct USB port, set the baud rate to 115200, and both software and hardware control to NONE.

The other way is to use the Linux 'screen' command, as shown below. It's much easier too!

```
screen /dev/ttyUSB1 115200
```

Also, in most cases, the virtual USB serial port is ttyUSB1. If it does not work, try ttyUSB0.

If all went well, you should be greeted with a familiar Angstrom login in the terminal as shown in Figure 2. The username for the same is 'root' and for the password, just press 'ENTER'. You should see the following prompt:

```
root@minnow:~#
```